Linear Control System Simulation: Analytical & Frequency Domain Approach

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1 Linear System Framework

1.1 Problem Statement

Flight control is a **linear time-invariant (LTI)** system for small perturbations:

$$\ddot{\theta}(t) = \frac{M(\delta(t))}{I_y} \tag{1}$$

where $M(\delta) = f(\delta)$ is aerodynamic torque, $\delta(t)$ is control deflection.

For small angles: $M(\delta) \approx K_M \delta$ where $K_M = \frac{dM}{d\delta}\big|_{\delta=0}$

1.2 State Space Representation

State vector: $\boldsymbol{x} = [\theta, \dot{\theta}]^T$

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}u, \quad y = \boldsymbol{C}\boldsymbol{x} \tag{2}$$

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 0 \\ K_M/I_y \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 1 & 0 \end{bmatrix} \tag{3}$$

1.3 PID Controller

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \dot{e}(t) \tag{4}$$

where $e(t) = \theta_{ref}(t) - \theta(t)$

Laplace domain:

$$U(s) = \left(K_p + \frac{K_i}{s} + K_d s\right) E(s) = C_{PID}(s) E(s) \tag{5}$$

2 Transfer Function Analysis

2.1 Open-Loop Plant

$$G(s) = \frac{\Theta(s)}{U(s)} = \frac{K_M/I_y}{s^2} \tag{6}$$

2.2 Closed-Loop System

$$H(s) = \frac{G(s) C_{PID}(s)}{1 + G(s) C_{PID}(s)} \tag{7}$$

Substituting:

$$H(s) = \frac{K_d s^2 + K_p s + K_i}{s^3 + (K_M K_d/I_y) s^2 + (K_M K_p/I_y) s + K_M K_i/I_y} \tag{8}$$

Characteristic equation:

$$s^3 + \alpha s^2 + \beta s + \gamma = 0 \tag{9}$$

where $\alpha = K_M K_d/I_y$, $\beta = K_M K_p/I_y$, $\gamma = K_M K_i/I_y$

3 Frequency Domain Solution

3.1 Fourier Transform Approach

For periodic reference $\theta_{ref}(t)$, decompose via Fourier series:

$$\theta_{ref}(t) = \sum_{k=-\infty}^{\infty} c_k e^{i\omega_k t}, \quad \omega_k = \frac{2\pi k}{T} \tag{10}$$

Response at each frequency:

$$\theta(t) = \sum_{k=-\infty}^{\infty} c_k H(i\omega_k) e^{i\omega_k t} \tag{11}$$

where $H(i\omega)$ is the frequency response.

3.2 Computational Advantages

Time domain: Solve ODE numerically (RK4, expensive)

Frequency domain:

1. FFT of reference signal: $\mathcal{F}\{\theta_{ref}\} = \tilde{\Theta}_{ref}(\omega)$
2. Multiply by transfer function: $\tilde{\Theta}(\omega) = H(i\omega)\tilde{\Theta}_{ref}(\omega)$
3. Inverse FFT: $\theta(t) = \mathcal{F}^{-1}\{\tilde{\Theta}(\omega)\}$

Complexity: $O(N \log N)$ vs $O(N \cdot N_{steps})$

4 Block Diagram Algebra

4.1 Cascaded Blocks

Servo dynamics (1st order):

$$G_{servo}(s) = \frac{1}{\tau_s s + 1} \tag{12}$$

Plant with servo:

$$G_{total}(s) = G_{servo}(s) \cdot G_{plant}(s) = \frac{K_M/I_y}{s^2(\tau_s s + 1)} \tag{13}$$

4.2 Delay Modeling

Pure time delay T_d:

$$G_{delay}(s) = e^{-sT_d} \approx \frac{1}{1 + sT_d} \quad \text{(Padé approximation)} \tag{14}$$

Total system:

$$G_{full}(s) = G_{delay}(s) \cdot G_{servo}(s) \cdot G_{plant}(s) \tag{15}$$

5 Analytical Solution for Special Inputs

5.1 Step Response

For $\theta_{ref}(t) = A \cdot u(t)$ (unit step):

$$\Theta(s) = H(s)\frac{A}{s} \tag{16}$$

Inverse Laplace (partial fractions):

$$\theta(t) = A\left(1 - \sum_{i=1}^{3} r_i e^{s_i t}\right) \tag{17}$$

where s_i are poles of $H(s)$, r_i are residues.

5.2 Sinusoidal Steady-State

For $\theta_{ref}(t) = A\sin(\omega t)$:

$$\theta_{ss}(t) = A|H(i\omega)|\sin(\omega t + \angle H(i\omega)) \tag{18}$$

No integration needed—direct evaluation!

6 Implementation Strategy

6.1 Hybrid Time-Frequency Algorithm

1. **Precompute** $H(i\omega)$ at discrete frequencies
2. **FFT** reference signal $\theta_{ref}(t)$
3. **Multiply** in frequency domain
4. **IFFT** to get response $\theta(t)$
5. **Refine** transients if needed (short time-domain integration)

6.2 Advantages Over Pure Time Integration

Property	Time Domain	Frequency Domain
Periodic inputs	$O(N^2)$	$O(N\log N)$
Nonlinear saturation	Native	Requires iteration
Multi-rate systems	Complex	Natural
Computational cost	High	Low

7 Nonlinearity Handling

Control saturation: $u(t) = \text{sat}(\delta, \delta_{max})$

Describing function method:

$$N(A) = \frac{4}{\pi A}\left[\delta_{max}\sin^{-1}\left(\frac{\delta_{max}}{A}\right) + \sqrt{A^2 - \delta_{max}^2}\right] \tag{19}$$

Replace linear gain with $N(A)$ for amplitude-dependent analysis.

8 Conclusion

Linear control theory enables:

- **Analytical** solutions for standard inputs
- **Frequency domain** fast computation via FFT
- **Modular** block composition (servo, delay, plant)
- **Systematic** PID tuning via pole placement

Engineering principle: Understand the mathematics → Design better controllers.